

Benchmarking the Recommendation Spectrum: An Empirical Analysis of Architectural Inductive Biases Across Randomized and Temporal Evaluation Regimes[★]

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Abstract

Modern recommender systems have transitioned rapidly from classical heuristic-driven algorithms to deep, graph-based, and sequential neural architectures. However, the performance of these models is deeply intertwined with their architectural inductive biases and the specific validation protocols used to evaluate them. In this work, we present a systematic, head-to-head empirical evaluation tracking the paradigm shift of recommendation methodologies from foundational models to state-of-the-art architectures. We evaluate eight distinct approaches spanning the entire evolutionary spectrum: Content-Based, Memory-Based Collaborative Filtering, Hybrid methods, Matrix Factorization, Neural Collaborative Filtering (NCF), Two-Tower architectures, LightGCN, and SASRec. To guarantee a robust architectural stress test, these models are rigorously benchmarked on the MovieLens-1M dataset across two distinct validation regimes representing fundamentally different operational scenarios: a standard 80/20 random split strategy and a strict, sequence-aware temporal leave-one-out (1-vs-all) evaluation protocol. Our findings expose critical trade-offs between architectural complexity and ranking robustness, demonstrating how model rankings dramatically shift under the influence of temporal data constraints versus randomized item allocation. We provide concrete insights into when lightweight, classical paradigms remain highly competitive against parameter-heavy neural counterparts.

1. Introduction

[★]Source Code Repository: <https://github.com/tuananhleuan/ml-rec-sys>